

Fusion-R · 多文档解码融合

示例元信息

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Synthetic Research Team · 2021

This is an invented demonstration document, not a published paper.

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3 Method

We extend Atlas-R by encoding each passage independently and fusing representations in the decoder.

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6 Limitations

Memory use grows with the number of retrieved passages. We leave adaptive passage selection to future work.

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4 Evaluation

At a fixed retrieval budget, independent passage encoding improves evidence coverage in our synthetic benchmark.

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Definition 1. In one normed vector space, a teaching fusion model is $h = \sum_i w_i h_i$, for finitely many nonnegative weights with $\sum_i w_i = 1$. This simplified model is not a description of a real decoder.

中文导读：在同一赋范向量空间中, 定义融合表示 $h = \sum w_i h_i$, 其中 $w_i \geq 0$ 且 $\sum w_i = 1$ 。

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Lemma 1. For nonnegative weights and vectors in the same normed vector space, $\text{norm}(\sum_i w_i (h_i - h^*)) \leq \sum_i w_i \text{norm}(h_i - h^*)$. Proof: apply the triangle inequality and positive homogeneity of the norm.

中文导读：对非负权重, 有 $\| \sum w_i (h_i - h^*) \| \leq \sum w_i \| h_i - h^* \|$ 。

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Theorem 1. Under Definition 1, if Assumption 1 holds, then $\text{norm}(h-h^*) \leq \epsilon$.

Proof: Definition 1 gives $h-h^* = \sum_i w_i (h_i-h^*)$. Lemma 1 bounds its norm by $\sum_i w_i \epsilon = \epsilon$, using Assumption 1. This does not guarantee factual accuracy or apply to arbitrary nonlinear decoders.

中文导读：若每个表示与目标 h^* 的距离至多为 ϵ ，则该加权融合表示与 h^* 的距离也至多为 ϵ 。

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Assumption 1. Every input representation satisfies $\text{norm}(h_i - h^*) \leq \epsilon$, with $\epsilon \geq 0$. This is an assumption, not a measured empirical guarantee.

中文导读：所有输入表示满足 $\|h_i - h^*\| \leq \epsilon$ ，其中 $\epsilon \geq 0$ 。

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Corollary 1. If all h_i equal h^* , the fused representation equals h^* . Proof: apply Theorem 1 with $\epsilon = 0$. A vector at norm distance zero equals h^* . This concerns the teaching vector model only.

中文导读：若所有输入表示都等于 h^* , 则融合表示也等于 h^* 。